

interpolated resolution? Still obvious? What if we told you that there are 11 scanners to choose from that cost just under Rs 6,000?

Specifications, prices, features and a whole lot of mumbo jumbo will only lead you to choices that are made out of confusion. But worry not! We have tested 23 scanners in three categories split by price, and have come up with the best scanners that you can choose from.

Digit Test Process

The scanners were categorised according to price as:

- **Budget scanners:** Up to Rs 6,000
- **Mid range scanners:** From Rs 6,001 to Rs 12,000
- **High-end scanners:** Rs 12,001 to Rs 20,000

The test system used to evaluate the scanners consisted of a Pentium III (Tualatin) 1.2 GHz processor, 256 MB SDRAM and a 40 GB, 7,200-rpm IBM Deskstar hard disk drive. Since the performance of the scanners would be affected by the interface it uses, we configured the parallel port to operate using the ECP+EPP mode. Where the scanner offered two modes of interfacing with the PC, we used the faster interface for testing. Hence, if the scanner supported Parallel and USB ports, we used the USB port and for those with USB and SCSI ports, we used SCSI. We used Photoshop 6.01 to evaluate the scanned images.

The operating system loaded was Windows 2000 Professional with DirectX 8.1 on a cleanly formatted system and the image of the drive with the operating system and all the basic drivers were loaded using Norton Ghost. This image was then restored after each scanner was tested so as to eliminate any interference from previously loaded TWAIN drivers.

Test methodology

We evaluated the scanners on various parameters such as features (30 per cent weightage), performance (40 per cent weightage), warranty and support (10 per cent weightage), and value for money (20 per cent weightage).

Features: We awarded points depending on the features and the capabilities of the scanners. These included the maximum resolution supported, document scan area, colour depth, and bundled accessories such as power adapters and interface cables. We also evaluated the software bundled with the scanner for basic image editing capabilities such as image scaling, colour and gamma correction, effect filters, batch scanning, and network scanning support. As beginners might be overwhelmed with the plethora of features that modern scanners offer, we checked the quality of documentation and manuals shipped with the scanners for quick setup instructions, a troubleshooting guide and after-sales support and contact information. This parameter was given a 30 per cent weightage in the overall score.

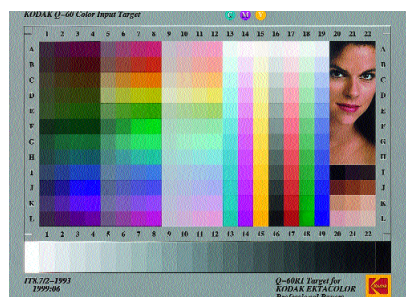
Performance: Each scanner was subjected to a range of tests that evaluated their performance on three basic criteria: scanning speed, quality and the ability to capture high-quality images. This parameter was given a 40 per cent weightage.

a) Scanning Speed: We scanned a text document and a colour photo in various modes at a constant 300 dpi. We scanned the text document in black and white mode and the photo was pre-scanned and then completely rescanned in true colour and

greyscale modes. The time taken to scan in each mode was noted using a stopwatch.

b) Image Quality: This consisted of two tests:

■ **The IT8 card test:** An IT8 card was used to determine the ability of a scanner to differentiate between dark, light and bright coloured areas (tonal variation). The IT8 card has three distinct areas (light, dark and bright), with three sub areas each. We took readings of the standard deviation values (using Photoshop 6.01)



of the red, blue and green channels of these three areas (including all the sub areas) and compared

The IT8 card tests a scanner's ability to pick up tonal variations in dark, light and bright areas of an image

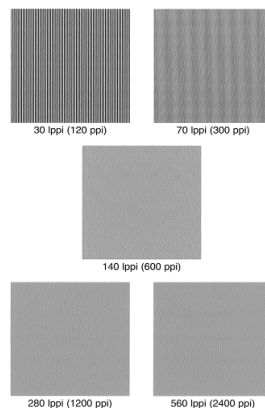
them against a reference value. A higher tonal deviation score in each zone indicates that the scanner can differentiate between the varying tones in a much better fashion. To arrive at the final score for each area, we took the average of all three readings for each sub area. Hence, 27 readings were taken in all, which on averaging came to nine different readings per area. We also measured the tonal deviations in greyscale mode by observing the tonal deviation of the various bars in the lower block of the IT8 card. Here we noted the maximum number of bars the scanner could distinctly identify.

■ **Photograph test:** A special glossy photograph was scanned at the highest optical resolution supported by the scanner. We compared the scanned image with the original photo for colour, overall smoothness of the image, realistic glossy effect as present in the original photograph, contrast, crispness in detail, etc.



The image used for the photo quality test

c) Resolution Test: To evaluate the resolution of each scanner we scanned a special document consisting of five patterned blocks in greyscale mode.



Each block contained closely packed lines with increasing density. The resolution of a scanner is judged by the number of distinct readings it can take per inch in each direction. Using Photoshop 6.01, we tried to select the individual lines present in the blocks. To differentiate between the lines, we used the Magic Wand tool in Photoshop.

The ability of the scanner to pick up closely packed lines in each of these blocks determines its resolution